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Impact of cardiac cycle and respiratory rhythm phase on visual attention in healthy young and older adults

Praghajieeth Raajhen Santhana Gopalan[1]✉, Jarmo Hämäläinen[1], Markku Penttonen[1] & Miriam S. Nokia[1]

Visual attention is crucial for human cognition, and previous research suggests that cardiorespiratory rhythms may modulate attentional processing. We investigated how cardiac and respiratory cycle phases influence visual attentional subprocesses: alerting and conflict-related processing. Healthy young and older adults ($n = 25$ per group) performed the Attention Network Test (ANT) while electroencephalography, respiration, and electrocardiography were recorded. We expected shorter reaction times (RTs) and larger event-related potential (ERP) amplitudes during inspiration and diastole compared to expiration and systole, respectively, with stronger effects in young adults. Behaviorally, young adults showed larger cue-related benefits than older adults, whereas conflict effects did not differ between groups. No reliable effects of cardiac or respiratory phase were observed on RTs. In contrast, ERP amplitudes revealed physiological modulation: respiratory phase produced task-dependent effects on N1 and P3 amplitudes, whereas cardiac phase exerted a robust global modulation, with larger amplitudes during late diastole compared to systole. These physiological effects did not differ between age groups, indicating preserved cardiorespiratory modulation of neural attentional processing in healthy aging. Future research should explore how individual differences in cardiorespiratory fitness and brain–body rhythm synchronization influence attention and learning.

Keywords Alerting, Executive control, Aging, Cardiac cycle, Breathing, Electroencephalogram

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[1]Department of Psychology and Centre for Interdisciplinary Brain Research, University of Jyväskylä, Jyväskylä, Finland. ✉email: praghajieeth.santhana@jyu.fi

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Author contributions
PRSG: Resources; Project administration; Conceptualization; Methodology; Investigation; Data curation; Formal analysis; Visualization; Writing – original draft; Writing – review & editing. JH: Conceptualization; Methodology; Writing – review & editing. MP: Conceptualization; Methodology; Writing – review & editing. MSN: Resources; Conceptualization; Supervision; Methodology; Writing – original draft; Writing – review & editing.

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Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to P.R.

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